HORUS UNIVERSITY – EGYPT

Advanced Statistics & probability

Lecture 3

By

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Sampling distribution of the variance (s^2)

- Let $X_1, X_2, \dots, X_n$ is a random sample od size n taken from normal population with mean μ and variance σ^2 then:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim {\chi^2}_{(n-1)}$$

- **Remark:**

✓ $E\left(\frac{(n-1)s^2}{\sigma^2}\right) = n-1 \Longrightarrow \frac{(n-1)}{\sigma^2} E(s^2) = n-1 \Longrightarrow E(s^2) = \sigma^2.$

✓ $V\left(\frac{(n-1)s^2}{\sigma^2}\right) = 2(n-1) \Longrightarrow \frac{(n-1)^2}{\sigma^4} V(s^2) = 2(n-1) \Longrightarrow V(s^2) = \frac{2\sigma^4}{(n-1)}.$

✓ $P(\chi^2 > a) = 1 - P(\chi^2 < a).$

✓ $P(a < \chi^2 < b) = P(\chi^2 > a)\text{-}P(\chi^2 > b)$

- **Example:**

Find the probability that a random sample of 25 observations from a normal population with variance 6, will have variance

a) **More than 9.1 .**

b) **Between 3.462 and 10.745.**

Solution

$$n = 25, \qquad \sigma^2 = 6$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

$$a) P(s^2 > 9.1) = P\left(\frac{(n-1)s^2}{\sigma^2} > \frac{9.1(25-1)}{6}\right) = P(\chi^2 > 36.4) = 0.05$$

$$b) P(3.462 < s^2 < 10.745) = P\left(\frac{3.462(25-1)}{6} < \frac{(n-1)s^2}{\sigma^2} < \frac{10.745(25-1)}{6}\right) = P(13.848 < \chi^2 < 42.98) =$$

$$P(\chi^2 > 13.848) - P(\chi^2 > 42.98) = 0.95 - 0.01 = 0.94$$

- **Example:**

If $X \sim N(10, 4)$, and $n = 8$, find $P(1.5 < s^2 < 6)$.

Solution

$$n = 8, \qquad \sigma^2 = 4$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim {\chi^2}_{(n-1)}$$

$$P(1.5 < s^2 < 6) = P\left(\frac{1.5(8-1)}{4} < \frac{(n-1)s^2}{\sigma^2} < \frac{6(8-1)}{4}\right)$$

$$= P(2.625 < \chi^2 < 10.5) = P(\chi^2 > 2.625) - P(\chi^2 > 10.5) = 0.84 - 0.08$$

$$= 0.76$$

Example:

A pharmaceutical company produces tablets that are assumed to have weights normally distributed with mean $\mu = 250\ mg$ and variance $\sigma^2 = 16mg^2$. To check the consistency of production, a quality control inspector randomly selects a sample of $n = 12$ tablets. Find the probability that the sample variance exceeds 25 mg²

Solution

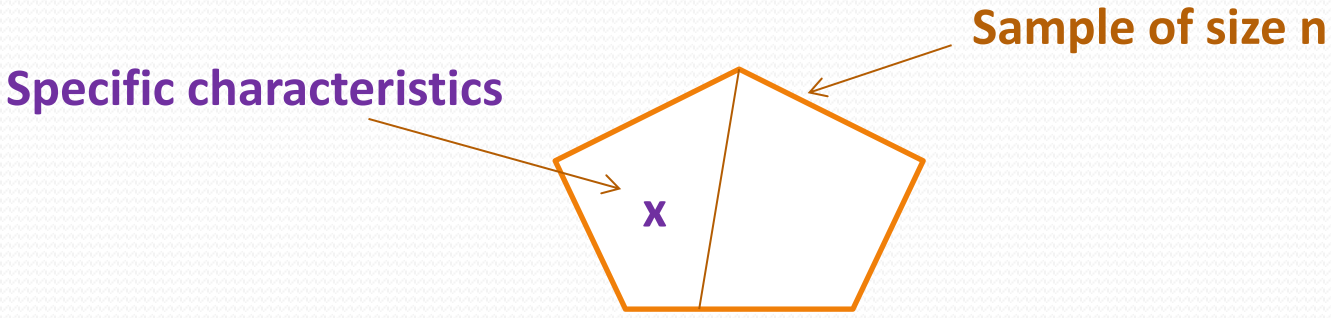
$$n = 12, \qquad \sigma^2 = 16$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

$$P(s^2 > 25) = P\left(\frac{(n-1)s^2}{\sigma^2} > \frac{25(12-1)}{16}\right) = P(\chi^2 > 17.1875) = 0.25$$

Sampling distribution of the proportion ($\hat{p}$)

- Let $X_1, X_2, \dots, X_n$ is a random sample od size n taken from population then:



The proportion is defined as:

$$\hat{P} = \frac{X}{n}, \qquad 0 \leq \hat{P} \leq 1.$$

Remark:

✓ $P \rightarrow$ proportion of the population.
✓ $\hat{P} \rightarrow$ proportion of the sample.

Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from population with proportion then:

$$E(\hat{P}) = P \qquad \text{and} \qquad V(\hat{P}) = \frac{P(1-P)}{n}$$

Then

$$Z = \frac{\hat{P} - P}{\sqrt{\frac{P(1-P)}{n}}} \sim N(0,1)$$

- **Remark:** We note that the quantity $\sqrt{\frac{P(1-P)}{n}}$ is called the standard error of the proportion.

Example:

A factory with 100 workers, including 20 smokers. If we select at random 36 workers, calculate the probability that the proportion of smokers in the sample is less than 0.1.

Solution

$$N = 100, \qquad X = 20 \rightarrow P = \frac{20}{100} = 0.2, \qquad\qquad n = 36$$

$$Z = \frac{\hat{P} - P}{\sqrt{\frac{P(1-P)}{n}}} \sim N(0,1)$$

$$P(\hat{P} > 0.1) = P\left(\frac{\hat{P} - P}{\sqrt{\frac{P(1-P)}{n}}} > \frac{0.1 - 0.2}{\sqrt{\frac{0.2(1-0.2)}{36}}}\right) = P(Z > -1.49) = 1 - \varphi(-1.49)$$

$$= 1 - \left(1 - \varphi(1.49)\right) = 1 - (1 - 0.9251) = 0.9251.$$

Example:

A recent study shows that **30%** of university students prefer online learning over traditional classroom instruction. A researcher randomly selects a sample of $n = 120$ students from the same university. Let $\hat{P}$ represent the sample proportion of students who prefer online learning. Answer the following questions:

a) Find the mean and standard deviation of the sampling distribution of $\hat{P}$.

b) Approximate the probability that more than **35%** of the sampled students prefer online learning.

Solution

$$P = 0.3, \qquad\qquad n = 120$$

$$Z = \frac{\hat{P} - P}{\sqrt{\frac{P(1-P)}{n}}} \sim N(0,1)$$

a) Mean of $\hat{P} = \mathrm{P} = 0.3,$ &&& standard deviation of $\hat{P} = \sqrt{\frac{0.3(1-0.3)}{120}} = 0.042$

b) $P(\hat{P} > 0.35) = P\left(\frac{\hat{P}-P}{\sqrt{\frac{P(1-P)}{n}}} > \frac{0.35-0.3}{\sqrt{\frac{0.3(1-0.3)}{120}}}\right) = P(Z > 1.19) = 1 - \varphi(1.19) = 1 - 0.8830$

$= 0.117.$

Example:

A health organization reports that 18% of adults are smokers in a certain city.

A public health researcher randomly selects a sample of $n = 200$ adults from the city. Let $\widehat{P}$ denote the sample proportion of smokers

a) **Find the mean and standard deviation of the sampling distribution of $\widehat{P}$.**

b) **Approximate the probability that the sample proportion is less than 15%.**

Solution

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Thank you for your Attention!